

SRM UNIVERSITY, ANDHRA PRADESH
SUMMER INTERNSHIP COURSE, June 2026
WEEKLY DIARY REPORT
(B.Tech. STUDENT BATCH 2024-28)



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Week 1 From 26-05-26 to 31-05-26		
S. No	Field	Answer
1	Project Title	Electrical Characterization of TSV Pinhole Defects Using RL/CG Models, HSPICE, Cadence, and HFSS
2	Project Description	The project aims to develop RLGC equivalent circuit models for oxide liner pinhole defects[in terms of RL and CG] in Through-Silicon Vias (TSVs). Mathematical equations are derived to represent defect behavior and implemented in HSPICE for signal integrity analysis including delay, noise, and crosstalk under in-phase and out-phase switching conditions. The models are further validated using Cadence and HFSS through S-parameter extraction and comparison. Results are

		analyzed and plotted to evaluate the impact of defect dimensions on TSV performance.
3	Outline of the Solution	1. Study TSV structures and pinhole defects. 2. Derive RLCG equivalent circuit model for DefectFree and Pinhole defect in terms of RL and CG models. 3. Implement the models in HSPICE, Cadence virtuoso, HFSS. 4. Analyze delay, noise, and crosstalk performance in hspice. 5. Extract S-parameters using Cadence. 6. Plot and analyze results using Origin. 7. Validate the models using HFSS simulations. 8. Compare circuit-level and electromagnetic simulation results.
4	Design of the Solution	The proposed solution models a TSV pinhole defect using two equivalent representations: an RL model and a CG model. Analytical equations are derived for the defect parameters and incorporated into circuit-level simulations. Performance metrics such as propagation delay, noise, and crosstalk are evaluated in HSPICE. S-parameter characterization is performed in Cadence and HFSS to validate the proposed models and assess their high-frequency behavior.
5	Hardware and Software Requirements to execute the project	Hardware: PC/Laptop. Software: Linux OS, HSPICE, Cadence Virtuoso, HFSS, WaveView, MATLAB (for equation verification and plotting), Text Editor.
6	Environment setup windows/linux/Raspberry pi/Arduino	Linux os
7	Concepts Used (Functions, header files, data types, and concepts (Loops, arrays, conditional statements, etc.). Explanation of the concepts). For Hardware projects also explain with respect to the code being developed	TSV structure, Pinhole defects, Resistance (R), Inductance (L), Capacitance (C), Conductance (G), Equivalent Circuit Modeling, Mathematical Equation Derivation, HSPICE Simulation, HFSS simulation, cadence virtuoso simulation, Signal Integrity Analysis, Delay Analysis, Noise Analysis, Crosstalk Analysis. Programming concepts include variables, equations, conditional statements, functions, and data processing.
8	Testing & Validation (Boundary tests and boundaries of inputs.	Deriving equations of pinhole defect modelled in terms of RL and CG, their equations simulations

	Possible inputs and corresponding outputs).	and results are compared to the existing Research paper works
9	Testing Material (Screenshots of working outputs. Images in case of hardware project	No simulation is done in week 01
10	User Manual	None
11	Technical Documentation	None
12	References	{1}Kim, J., Pak, J. S., Cho, J., Song, E., Cho, J., Kim, H., ... & Kim, J. (2011). High-frequency scalable electrical model and analysis of a through silicon via (TSV). IEEE Transactions on Components, Packaging and Manufacturing Technology, 1(2), 181-195. {2}Chandrakar, M., Solanki, K., Majumder, M. K., & Sharma, R. Reliability Concerns in Through Silicon Via based 3D Integration: Fabrication to Packaging. {3}Kirtonia, P., Williams, S., Akter, S., Bayoumi, M., & Khalil, K. (2025). A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs. IEEE Transactions on Circuits and Systems I: Regular Papers. {4} Liu, C., Dong, G., Zhi, C., & Zhu, Z. (2025). Modelling and simulation of TSV considering void and leakage defects. Journal of Computational Electronics, 24(2), 61. {5}Arumí, D., Rodríguez-Montañés, R., & Figueras, J. (2015). Prebond testing of weak defects in TSVs. IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 24(4), 1503-1514.
13	Daily Work Breakdown	
	Day 1	Notes: Conceptual understanding of types of Capacitances in the TSV
	Day 2	Notes Research paper reading and conceptual understanding of Resistance, Inductance and conductance Equation modelling in a Through Silicon Via

	Day 3	Using the reference equations of the defect free TSV from the reference paper Started to model our proposed Defect free TSV physical structure.
	Day 4	Designing Electrical Equivalent Model for designed physical Defect Free TSV model.
	Day 5	Deriving R,L,G,C equations of the Electric equivalent quantities.